VIGNESH SUGUNATHAN

EMBEDDED SYSTEMS | FIRMWARE DEVELOPMENT | HARDWARE INTEGRATION

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in embedded firmware development, microcontroller programming, hardware-firmware integration, circuit debugging, peripheral interfacing and system validation. Proficient in Embedded C/C++, STM32, PIC16F877A, ESP32, Arduino, UART, SPI, I2C, GPIO, ADC, PWM, timers, interrupts and EEPROM. Experienced in developing and testing embedded systems for robotics, IoT, automation and real-time monitoring applications.

TECHNICAL SKILLS

Programming	Embedded C, C, C++, Python
Microcontrollers	STM32F103C8T6, ARM Cortex-M3, PIC16F877A, ESP32, ATmega328P, MSP430, Arduino
Protocols	UART, SPI, I2C, LoRa, Wi-Fi, Bluetooth
Peripherals	GPIO, ADC, PWM, Timers, Interrupts, RTC, EEPROM
Hardware & Firmware	Firmware Development, Peripheral Bring-up, Hardware-Firmware Integration, Sensor Interfacing, Circuit Debugging, Prototyping, Functional Testing, System Validation
Tools	STM32CubeIDE, MPLAB X, Arduino IDE, Proteus, KiCad, Git, VS Code, Linux

PROFESSIONAL EXPERIENCE

Robotics Instructor | NxtWave | September 2026 – Present | India

• Conduct practical robotics and IoT learning sessions for university students.

• Guide learners through hands-on projects involving electronics, sensors, microcontrollers and programming.

• Explain technical concepts, troubleshoot project issues and support students during implementation.

AI Robotics Engineer | Centrado Tech Solutions Pvt. Ltd. | December 2025 – August 2026 | Chennai, India

• Developed and debugged embedded firmware for robotics, IoT-enabled STEM products and microcontroller-based development boards.

• Integrated microcontrollers, sensors, actuators and wireless communication modules for embedded prototypes.

• Performed hardware bring-up, firmware debugging, functional validation and system-level testing.

• Collaborated with electronics and software teams during product development, design reviews, troubleshooting and demonstrations.

Embedded Systems Intern | Emertxe Information Technologies | January 2025 – February 2025 | Chennai, India

• Developed Embedded C firmware for PIC16F877A using GPIO, timers, ADC, interrupts and EEPROM.

• Debugged and validated firmware-hardware interfaces during peripheral bring-up and functional testing.

• Developed and tested a Microwave Oven Automation System using embedded control logic and microcontroller peripherals.

SELECTED PROJECTS

STM32 Environmental Monitoring & Display System — STM32F103C8T6, Embedded C, I2C, UART
Integrated DHT11, DS1307 RTC and SH1106 OLED using I2C; implemented UART debugging, sensor acquisition, modular firmware and functional validation.

Microwave Oven Automation System — PIC16F877A, Embedded C, MPLAB X, Proteus
Developed appliance-control firmware using GPIO, timers, ADC, interrupts and EEPROM; simulated, debugged and validated the system in Proteus.

Autonomous Surveillance & Hazard Detection Robot — ESP32, Embedded C, PWM, IoT
Integrated obstacle detection, hazard sensing, PWM motor control and IoT-based remote monitoring.

Solar-Powered LoRa Mesh Communication Network — Arduino Nano, ESP8266, SX1278, AES-128
Developed encrypted LoRa communication with RSSI-based routing for emergency messaging; recorded approximately 96% packet delivery across 500 transmissions up to 200 m during outdoor testing.

EDUCATION & ACHIEVEMENTS

B.E. Electronics and Communication Engineering — Agni College of Technology, Chennai | 2022–2026 | CGPA: 8.2 / 10
Runner-Up, SRM National Level Robotics Hackathon, 2025. Published two IEEE research papers in Image Processing and Machine Learning.